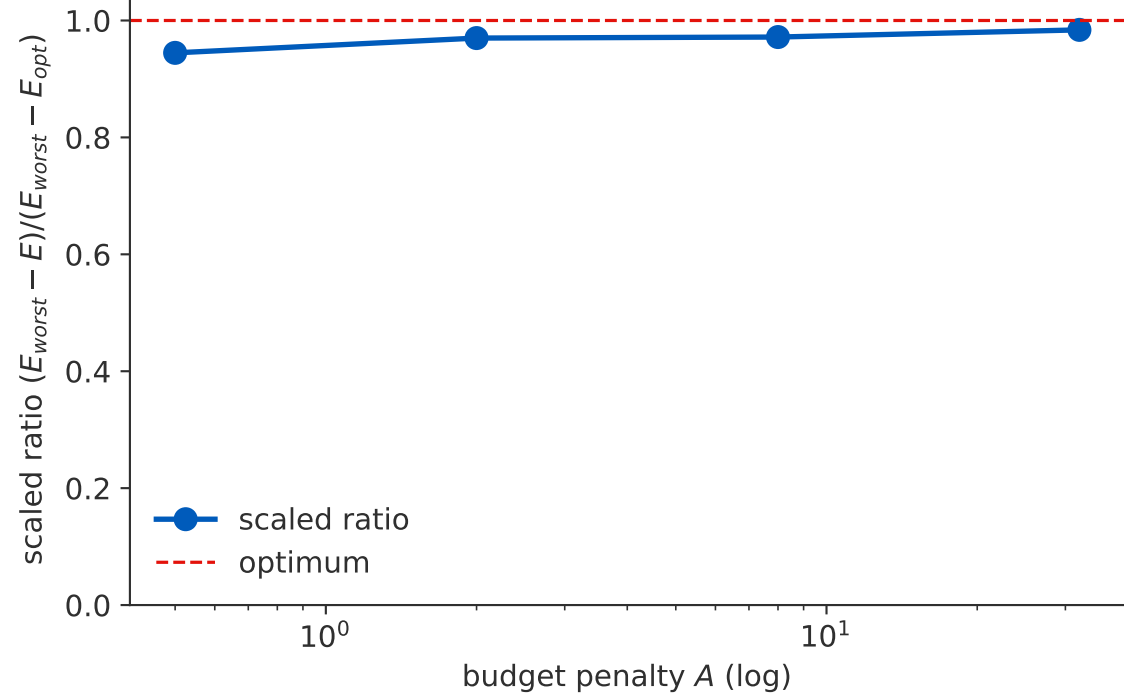


QAOA approximation ratio vs penalty

expect monotone improvement as A grows



Measurement concentration

higher = penalty is doing its job

